However, AI adoption here is playing out at two speeds.

- ***The government track:*** The public sector has moved decisively. The national AI strategy has been laid out. Mission-mode projects in agriculture, healthcare, and education are underway. State-level initiatives are exploring AI for crop yield forecasting, traffic management, and multilingual citizen services.
- ***The enterprise track:*** Many private sector companies are still cautious. Proof-of-concept projects abound, but full-scale production deployments are rare. Concerns about cost, skill gaps, integration complexity, and unclear ROI slow momentum.

Caution in this era is a competitive disadvantage. AI-native companies can innovate faster, operate leaner, and personalise at scale. Once those dynamics set in, catching up with them is exponentially harder.

The semiconductor foundation

Every AI breakthrough you read about, be it ChatGPT generating coherent essays, Stable Diffusion creating photorealistic art, or DeepMind solving protein folding, rests on advances in semiconductor technology.

The journey began with general-purpose CPUs in the early days. Over time, the need for highly parallel processing, particularly for graphics and AI workloads, led to the rise of GPUs. Today, specialised AI accelerators handle the massive tensor operations required for training large neural networks.

Intel's strategy is not to champion one architecture over another, but to enable AI everywhere:

- CPUs optimised with advanced vector extensions (AVX512) and advanced matrix extensions for AI workloads.
- Discrete GPUs designed for deep learning.
- Dedicated AI accelerators for high-efficiency inference.

Coupled with software like OpenVINO, Intel makes it possible to run AI on everything from a low-power edge device in a rural clinic to a hyperscale data centre handling petabytes of training data.

Just as the skill of the driver in Formula 1 (AI algorithms) is limited by the engine's capabilities (hardware), the next leaps in AI performance will come as much from semiconductor innovation as from algorithmic advances.

From generative to agentic AI: Real-world examples

A generative AI tool in retail can produce product descriptions for an e-commerce site based on bullet points from the merchandising team.

An agentic AI, in contrast, can:

- Pull sales data from the last quarter.
- Identify which product categories need a marketing push.
- Create tailored descriptions for those products.
- Generate accompanying social media creatives.
- Schedule posts based on historical engagement data.
- Monitor performance and adjust the campaign in real time.

In manufacturing, generative AI may draft a maintenance manual. Agentic AI could monitor sensor data across a factory, predict equipment failures, schedule repairs, order replacement parts, and update manuals dynamically.

Key Indian AI use cases emerging today

Agriculture: Satellite imagery, IoT sensors, and weather models combine through AI to predict yields, detect pest infestations early, and advise farmers on optimal planting schedules.

Healthcare: AI-assisted diagnostics; telemedicine platforms that triage patients based on urgency; and personalised treatment recommendations using patient history.

Education: Adaptive learning platforms that adjust difficulty level based on student progress, available in multiple Indian languages.

Public services: Automatic translation of government communications into 22 official languages; AI chatbots handling citizen queries at scale.

Each of these not only improves efficiency but extends reach, which is critical in a country of 1.4 billion where resources are often stretched.

The role of open source in AI democratisation

The future of AI cannot be monopolised by a handful of tech giants. Open source frameworks and tools ensure that innovation is distributed, talent pipelines are broad, and localised solutions can emerge.

Intel contributes actively to open source AI ecosystems optimising PyTorch, TensorFlow, and integrating with repositories like Hugging Face. The company's OpenVINO toolkit allows developers to deploy models across CPUs, GPUs, and accelerators without rewriting code.

Through techniques like model quantisation and pruning, Intel makes it possible for powerful AI models to run on modest hardware. This is essential for rural, edge, and embedded applications where high-end GPUs are impractical.

Security, safety, and trust

As AI systems become more capable, the stakes of misuse, whether intentional or accidental, rise. Intel addresses this in multiple layers.

Hardware root of trust: Secure boot and encryption to protect data even while in use.

Confidential computing: Workloads are isolated in secure enclaves, shielding sensitive data.